

Gallbladder Disease: Biliary Colic And Cholecystitis

Definition

The spectrum of gallbladder disease ranges from asymptomatic gallstones to biliary colic, cholecystitis, choledocholithiasis and cholangitis.

Cholecystitis is inflammation of the gallbladder caused by obstruction of the cystic duct, usually by a gallstone (calculous cholecystitis). The inflammation may be sterile or bacterial. The obstruction may be acalculous or caused by sludge.

Choledocholithiasis occurs when the stones become lodged in the common bile duct; from this, cholangitis and ascending infections can occur.

Causes

Biliary Colic

Gallstones temporarily obstruct the cystic duct or pass into the common bile duct.

Cholecystitis

The cystic duct or common bile duct becomes obstructed for hours, or gallstones irritate the gallbladder. Bacterial infection is thought to be a consequence, not a cause, of cholecystitis.

The most common organisms are *E. coli*, *Klebsiella* spp. and enterococci. Stones of the common bile duct (occurring in 10% of patients with gallbladder disease) are secondary (from the gallbladder) or primary (formed in the bile ducts).

Risk Factors

The phrase "fair, fat and fertile female" summarizes the major risk factors for gallstones. Although gallstones and cholecystitis are more common in women, men with gallstones are more likely to experience cholecystitis than women with gallstones. It is unknown if women who are pregnant or have multiple pregnancies are more likely to have gallstones or if they simply have more symptoms of the stones.

Some oral contraceptives and estrogen replacement therapy may increase the risk of gallstones.

Rates of gallstones, cholecystitis and stones of the common bile duct increase with age. Elderly clients are more likely to have asymptomatic gallstones that result in serious complications without gallbladder colic.

The causes of gallstones in teenagers are the same as for adults, and there is a higher prevalence among girls and during pregnancy.

History

Most gallstones (60% to 80%) are asymptomatic. Small stones are more likely to be symptomatic than large ones. Almost all patients experience symptoms before complications occur. Indigestion, belching, bloating and intolerance of fatty food are thought to be typical symptoms of gallstones; however, these symptoms are just as common in people without gallstones and frequently are not cured by cholecystectomy.

Biliary Colic

- 1-5 hours of constant pain, commonly in the epigastrium or right upper quadrant
- Pain may radiate to the right scapular region or back
- Client tends to move around to seek relief from pain
- Onset of pain occurs hours after a meal, frequently at night, waking the client from sleep
- Peritoneal irritation by direct contact with the gallbladder localizes the pain to the right upper quadrant
- Pain is severe, dull, or boring and constant (not colicky)
- Associated symptoms include nausea, vomiting, pleuritic pain and fever

Cholecystitis

- Persistence of the biliary obstruction leads to cholecystitis
- Persistent right upper quadrant pain
- The character of the pain is similar to the pain associated with gallbladder colic, except that it is prolonged and lasts for hours or days
- Nausea, vomiting and low-grade fever are more commonly associated with cholecystitis

Physical Findings

- Vitals signs parallel the degree of illness
- Clients with biliary colic have relatively normal vital signs
- Clients with cholangitis are more likely to have tachycardia or hypotension (or both) and fever
- Fever may be absent, especially in elderly clients
- Jaundice (in < 20% of patients)

Abdominal Examination in Gallbladder Colic and Cholecystitis

- Epigastric or right upper quadrant tenderness
- Murphy's sign (an inspiratory pause on palpation of the right upper quadrant; specific but not sensitive for gallbladder disease)
- Guarding on palpation
- Fullness in the right upper quadrant may be palpated

As in anyone with abdominal pain, a complete physical examination must be performed (including rectal and pelvic examinations in women). In elderly and diabetic clients, occult cholecystitis or cholangitis may be the source of fever, sepsis or changes in mental status.

Differential Diagnosis

- Appendicitis
- Acute bowel obstruction
- Ascending cholangitis
- Cholelithiasis
- Diverticular disease
- Gastroenteritis
- Hepatitis
- Inflammatory bowel disease
- Mesenteric ischemia
- Myocardial infarction
- Pancreatitis
- Bacterial pneumonia
- Eclampsia
- Hyperemesis gravidarum
- Urinary tract infection
- Renal calculi

Complications

Biliary Colic

- Cholecystitis

Acute Cholecystitis

- Perforation
- Gangrene
- Peritonitis
- Cholangitis
- Abscess
- Fistula
- Pancreatitis
- Ileus

Diagnostic Tests

The choice of laboratory tests will depend on whether the client is well enough to be treated as an outpatient or requires admission to hospital. The results of lab tests should be completely normal if the client has cholelithiasis or gallbladder colic.

- White blood cell (WBC) count and liver function tests (LFTs) (AST, ALT, bilirubin and alkaline phosphate levels) may be helpful in the diagnosis of cholecystitis
- An elevated WBC count is expected; however, a normal value does not rule out cholecystitis
- Bilirubin >3.5 µmol/L may indicate stone in the common bile duct or ascending cholangitis
- Mild elevation of amylase (up to 3 times normal level) may be present in cholecystitis, especially if there is gangrene
- Urinalysis
- Pregnancy test for women of childbearing age

Management Of Biliary Colic

Goals of Treatment

- Relieve pain, nausea and vomiting
- Prevent complications

Appropriate Consultation

Consult physician if pain does not resolve, if fever develops or if significant vomiting continues, as these symptoms indicate that a complication may be developing.

Nonpharmacologic Interventions

- Bed rest
- Clear fluids if vomiting

Client Education

- Explain disease process and prognosis
- Counsel client about appropriate use of medications (dose, frequency)
- Recommend low-fat food as tolerated, once pain resolves

Pharmacologic Interventions

Analgesia

Primary pain should be controlled with anticholinergic antispasmodics:

hyoscine butylbromide (B class drug),

10 mg IM q6h prn (max 100mg/day)

hyoscine butylbromide (C class drug),

10mg, 1-2 tabs, PO q6h (max 6 tabs/day)

Secondary pain should be controlled with meperidine; **do not use morphine**, which may increase tone in the Oddi's sphincter:

meperidine (D class drug), 50-100 mg IM q3-4h prn

Consult physician for IV order

Antiemetics to relieve vomiting and nausea:

dimenhydrinate (A class drug), 25-50 mg IM q4-6h prn

Monitoring and Follow-Up

Monitor for a few hours. When nausea and vomiting have resolved, push clear fluids.

Follow-up in 24 hours is recommended. If pain increases, fever develops, or the client is unable to tolerate intake by mouth because of vomiting, manage as for acute cholecystitis.

Management Of Cholecystitis

Goals of Treatment

- Relieve pain, nausea and vomiting
- Prevent complications

Appropriate Consultation

Consult physician if pain does not resolve, if fever develops or if significant vomiting continues indicating that a complication may be developing.

Adjuvant Therapy

For clients with severe pain prehospital care should include the following:

- Oxygen, if client is unstable on presentation
- IV therapy with normal saline, rate adjusted according to age, state of hydration and pre-existing medical problems

Nonpharmacologic Interventions

- Bed rest
- Nothing by mouth

Pharmacologic Interventions

Analgesia

Pain control should be given early, without waiting for the diagnosis or surgical consult.

Primary pain control should be with anticholinergic antispasmodics:

hyoscine butylbromide (B class drug),

10 mg IM q6h prn (max 100mg/day)

Secondary pain control should be with meperidine; do not use morphine, which may increase tone in the Oddi's sphincter:

meperidine (D class drug), 50-100 mg IM q3-4h prn

Antiemetics

dimenhydrinate (A class drug), 25-50 mg IM q4-6h

Meperidine and dimenhydrinate can be mixed in the same syringe, but should be used immediately.

Antibiotics

For mild cholecystitis, where inflammation is the primary process, antibiotics are not usually used.

For acute cholecystitis (if client is febrile and acutely ill), draw a blood sample for culture and consult physician for IV antibiotics.

For clients with allergy to penicillin use only metronidazole.

Monitoring and Follow-Up

Monitor pulse oximetry, vital signs (frequent), blood glucose, intake and output.

Severe cholecystitis can evolve into sepsis or cholangitis, especially in diabetic or elderly clients in whom the diagnosis may be delayed.

Referral

Medevac as soon as possible; surgical consult is required.